

Title

Antenatal Physical Activity as a Preventive Strategy Against Postpartum Mood Disorder: A Randomized Controlled Trial Protocol

Abstract

Postpartum depression affects a substantial proportion of women and impairs maternal-infant bonding and family functioning. Physical activity is known to influence mood regulation and stress resilience. This protocol describes a single-center randomized trial designed to test whether a structured aerobic exercise regimen initiated during early pregnancy and maintained through mid-gestation reduces the incidence of postpartum depression in women with uncomplicated pregnancies. Approximately 400 pregnant women will be enrolled at the first prenatal visit and randomly assigned to either participate in a supervised aerobic exercise program (three 60-minute sessions per week) or continue with standard prenatal care. The primary outcome is postpartum depressive symptom severity at three months postpartum, measured by validated screening instruments. Secondary outcomes include clinical diagnosis rates, neonatal outcomes, and maternal physical and metabolic parameters. We hypothesize that maintained prenatal aerobic activity reduces postpartum depression through multiple pathways including improved mood regulation and stress coping.